

Matrices of Linear Transformations

Geometric Algorithms
Lecture 8

Practice Problem

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 - 3x_1 \\ 9x_1 - 3x_2 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Is $\mathbf{b}$ in the range of T ? If so, determine $\mathbf{v}$ such that $T(\mathbf{v}) = \mathbf{b}$

$$\begin{bmatrix} x_2 - 3x_1 \\ 9x_1 - 3x_2 \end{bmatrix} = T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x_2 - 3x_1 \\ 9x_1 - 3x_2 \end{bmatrix} = T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

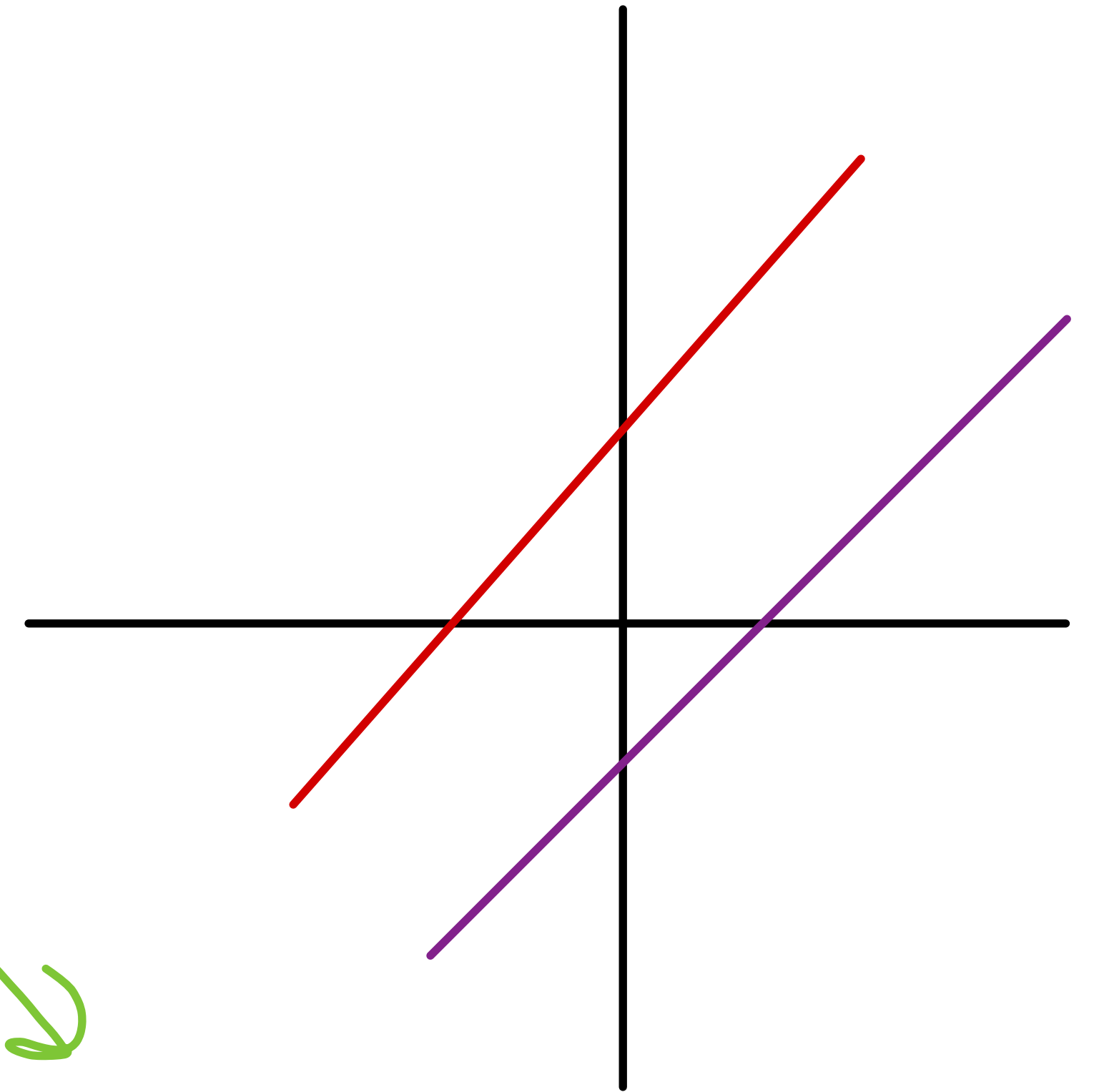
$$-3x_1 + x_2 = 2$$

$$9x_1 - 3x_2 = -1$$

$$\begin{bmatrix} -3 & 1 & 2 \\ 9 & -3 & -1 \end{bmatrix} \sim \begin{bmatrix} -3 & 1 & 2 \\ 0 & 0 & 5 \end{bmatrix}$$

$$0 = 5$$

$$\therefore \begin{bmatrix} 2 \\ -1 \end{bmatrix} \notin \text{ran}(T)$$



$$\begin{bmatrix} -3 \\ 9 \end{bmatrix} x_1 + \begin{bmatrix} 1 \\ -3 \end{bmatrix} x_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\text{span} \left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix} \right\} \not\ni \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Keywords

matrix of a linear transformation

standard basis vectors (standard coordinate vectors)

2D linear transformations

the unit square

one-to-one

onto

Geometry of Matrix Transformations

Motivating Questions

What kind of functions can we define this way?

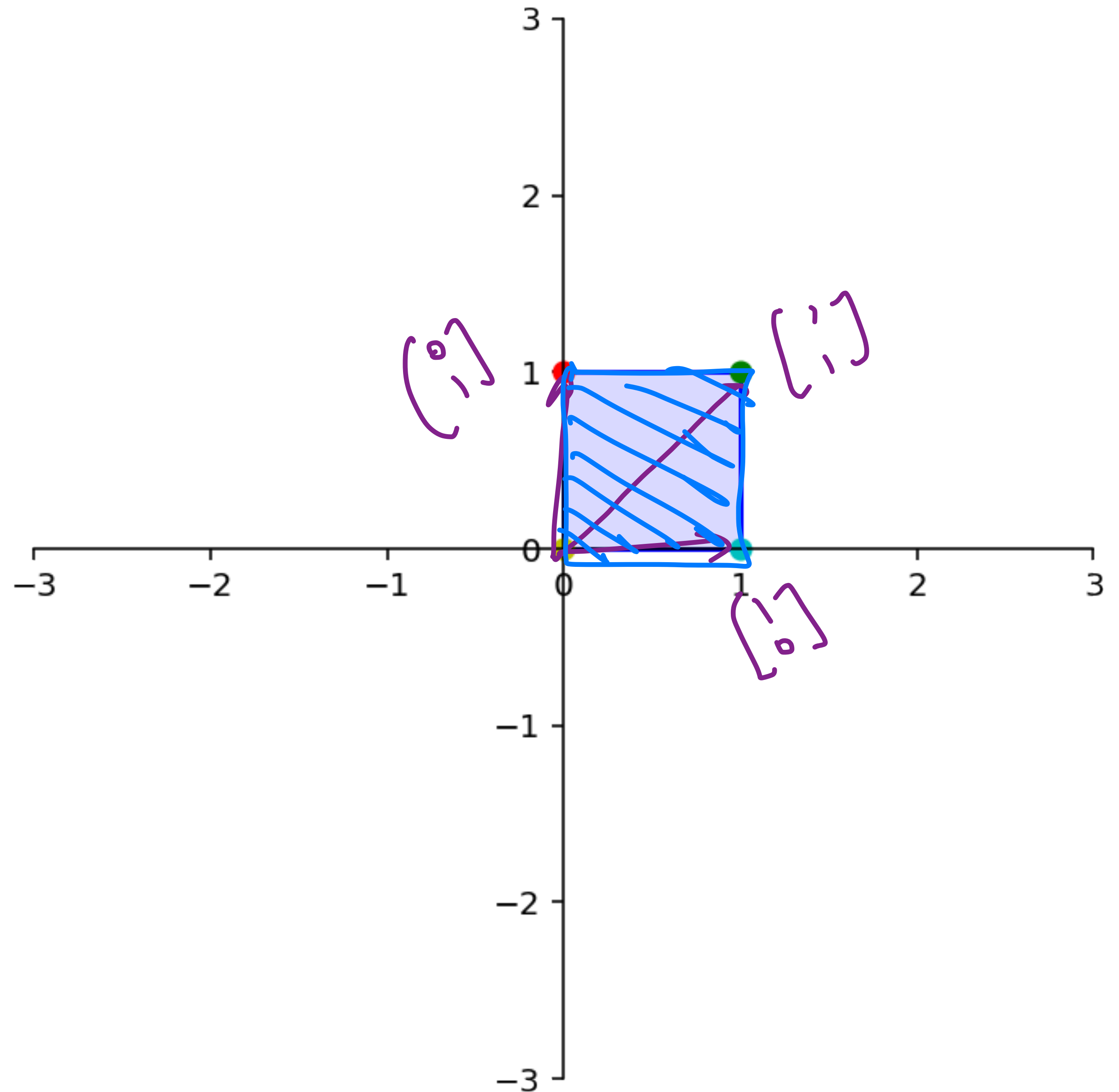
How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

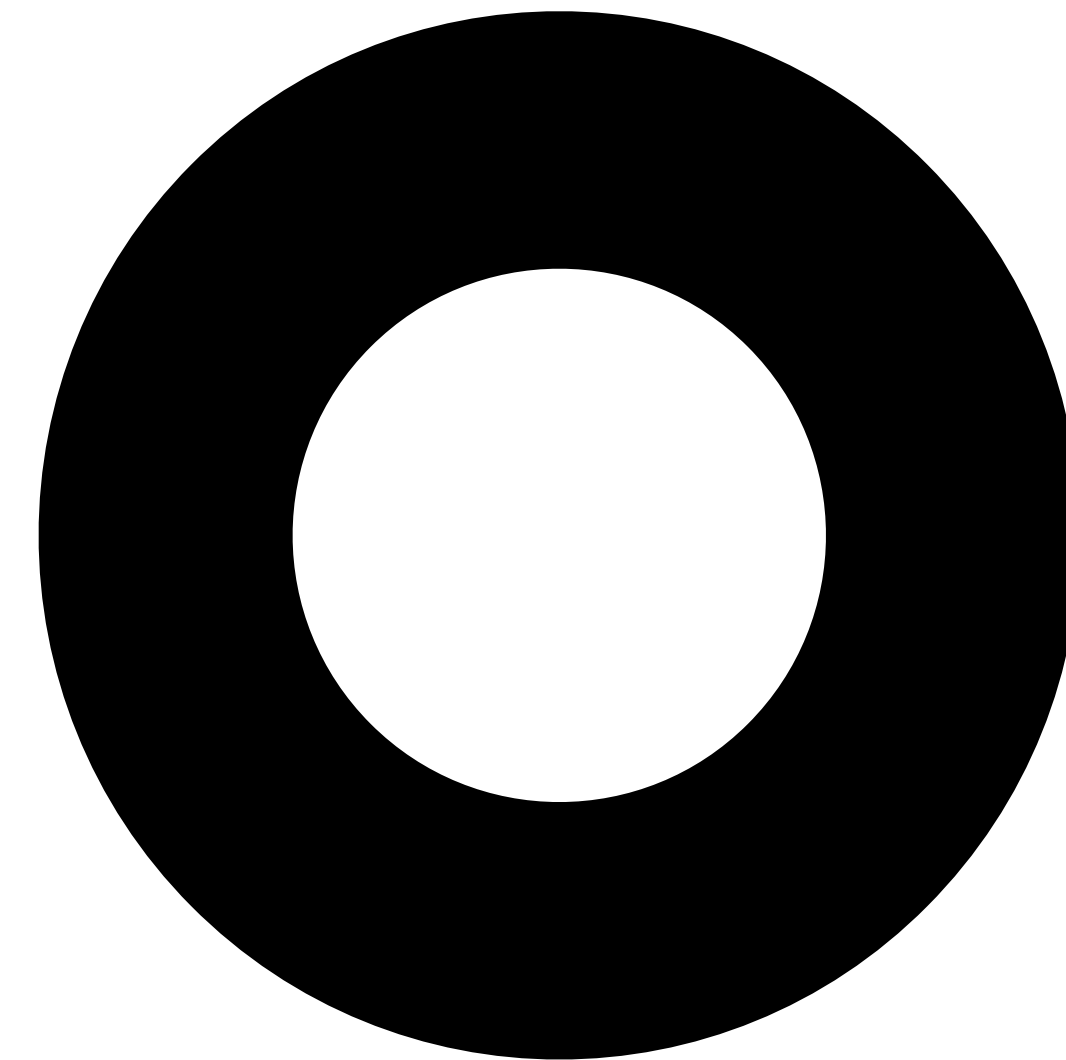
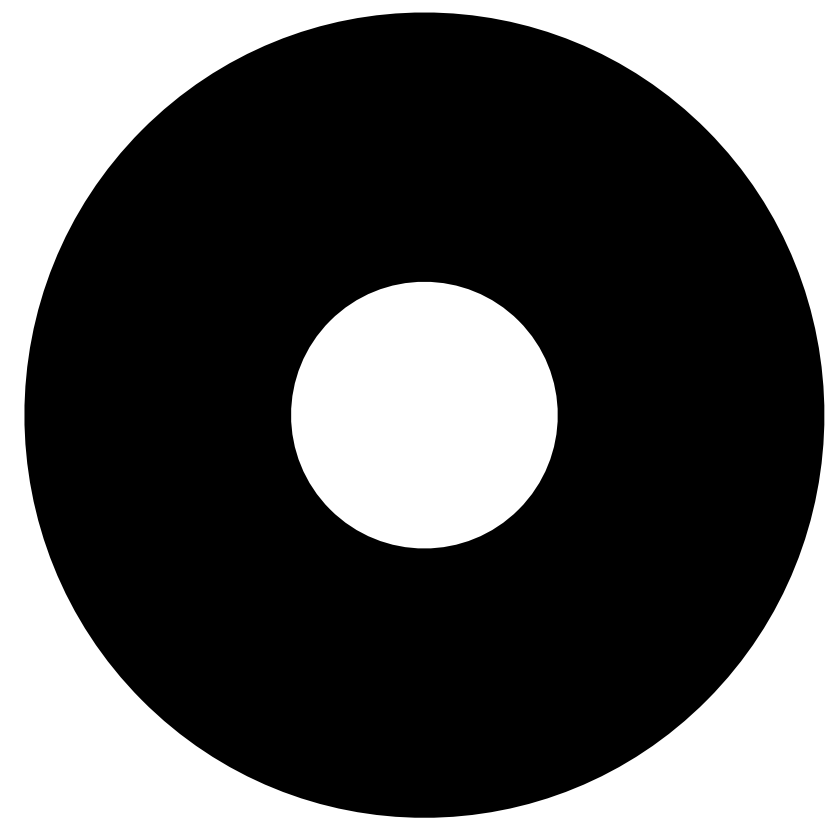
Matrix transformations change the
"shape" of a set of set of
vectors (points)

The Unit Square

The *unit square* is the set of points in $\mathbb{R}^2$ enclosed by the points $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$.



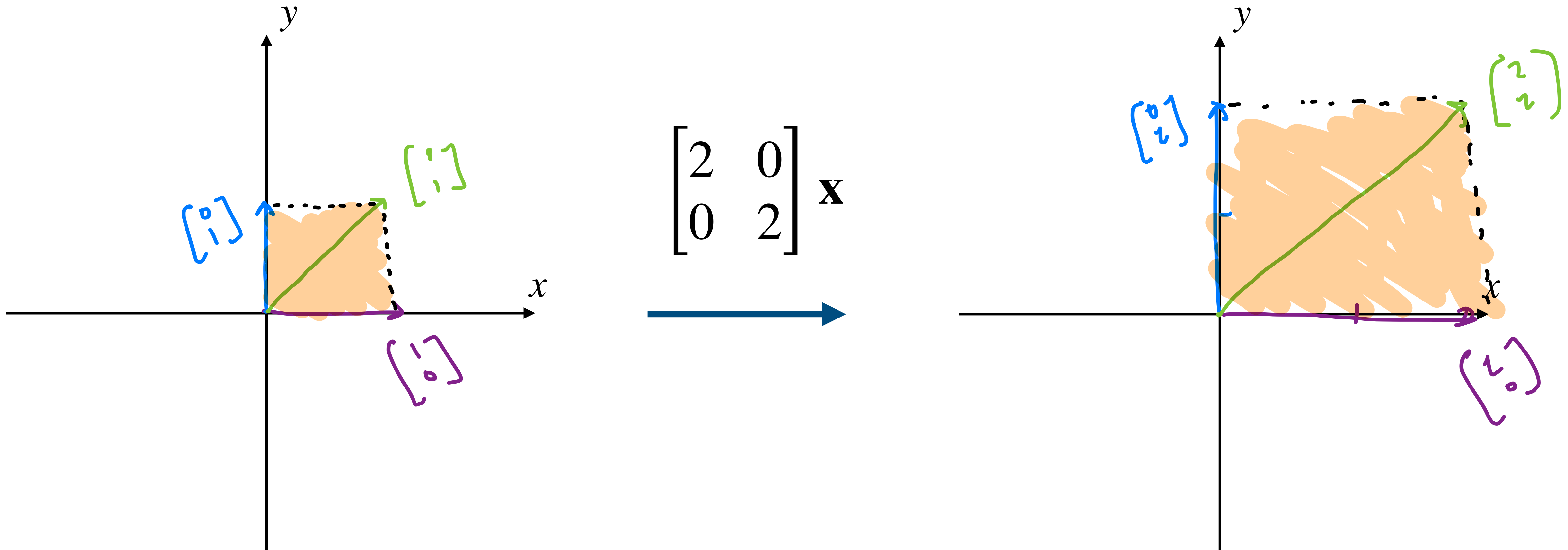
Example: Dilation



$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$$

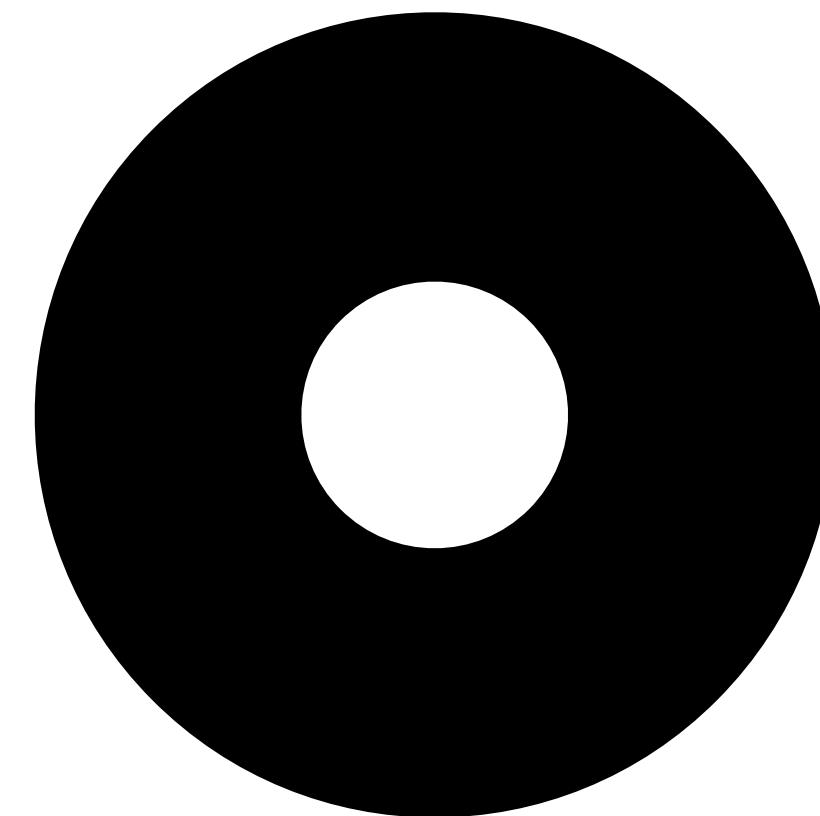
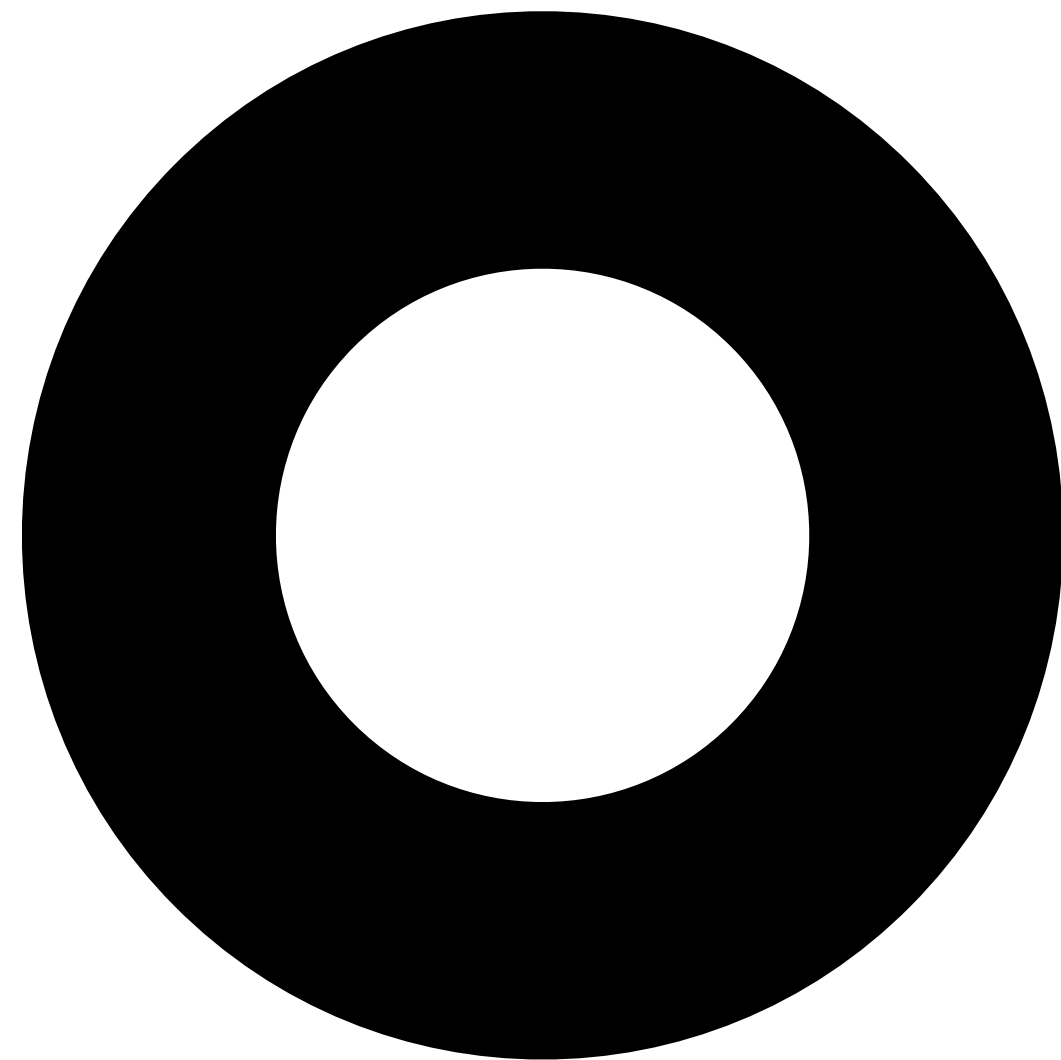
Example: Dilation

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$



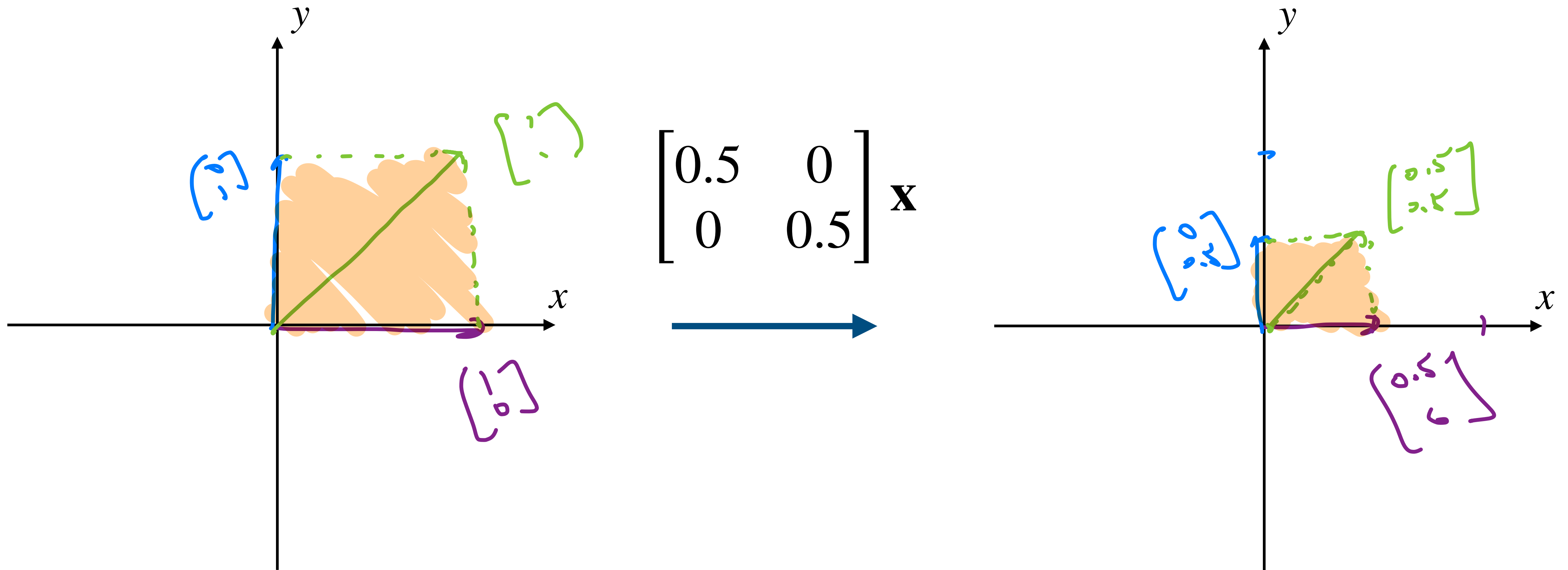
if $r > 1$, then the transformation pushes points away from the origin

Example: Contraction



Example: Contraction

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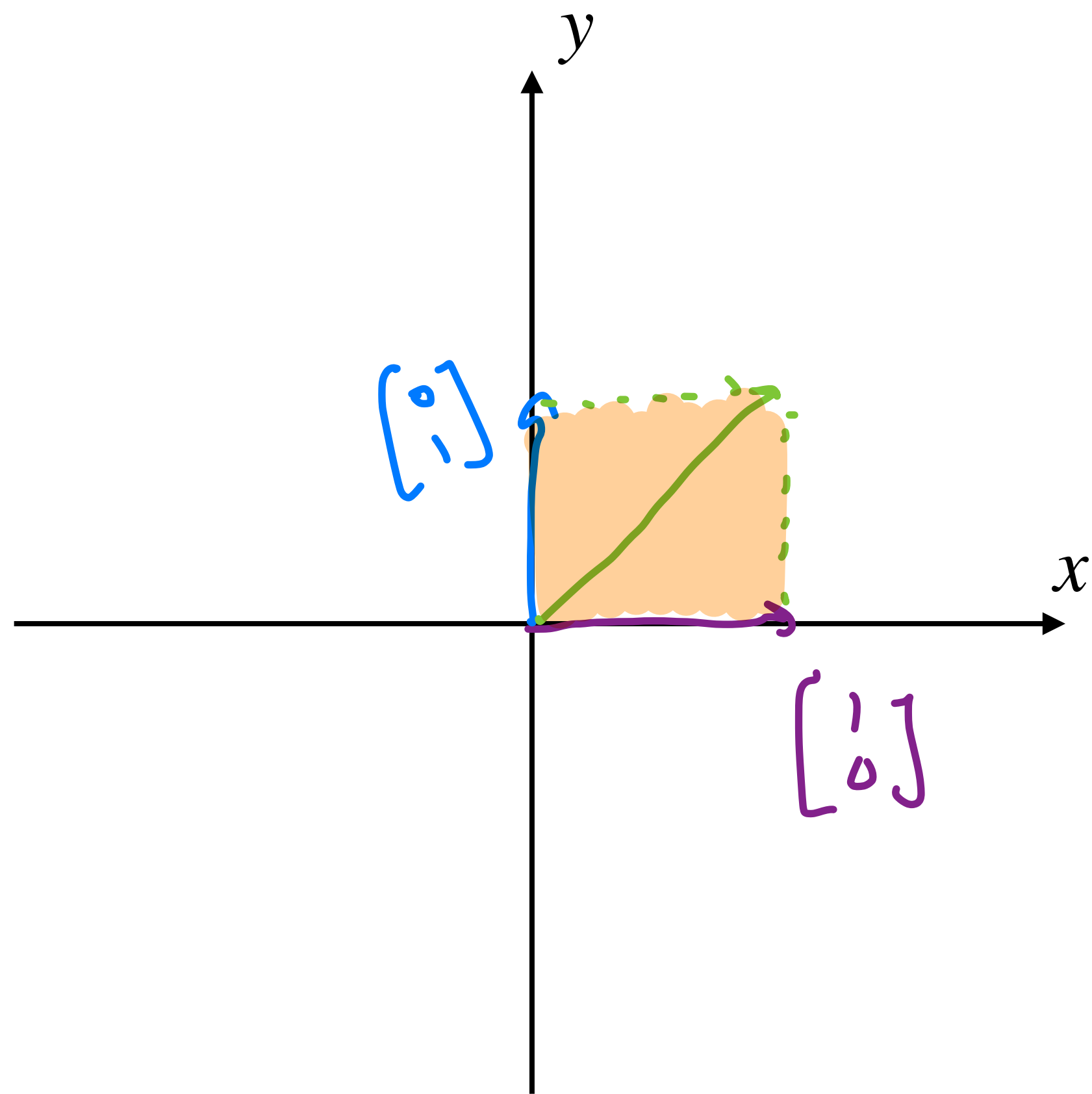
if $0 \leq r \leq 1$, then the transformation pulls points towards the origin

Example: Shearing



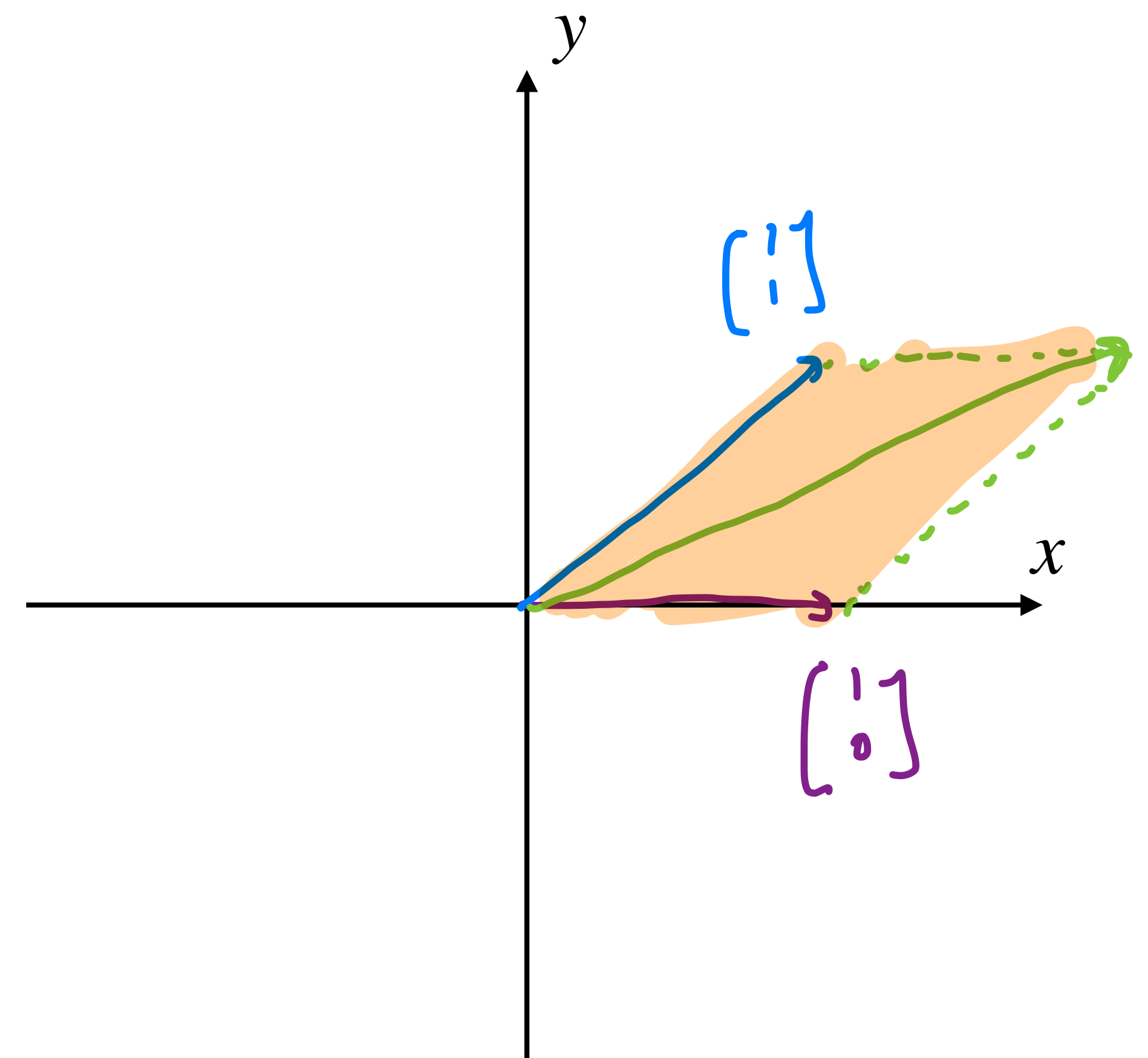
Example: Shearing

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$

A blue arrow points from the first diagram to the second diagram, indicating the transformation.

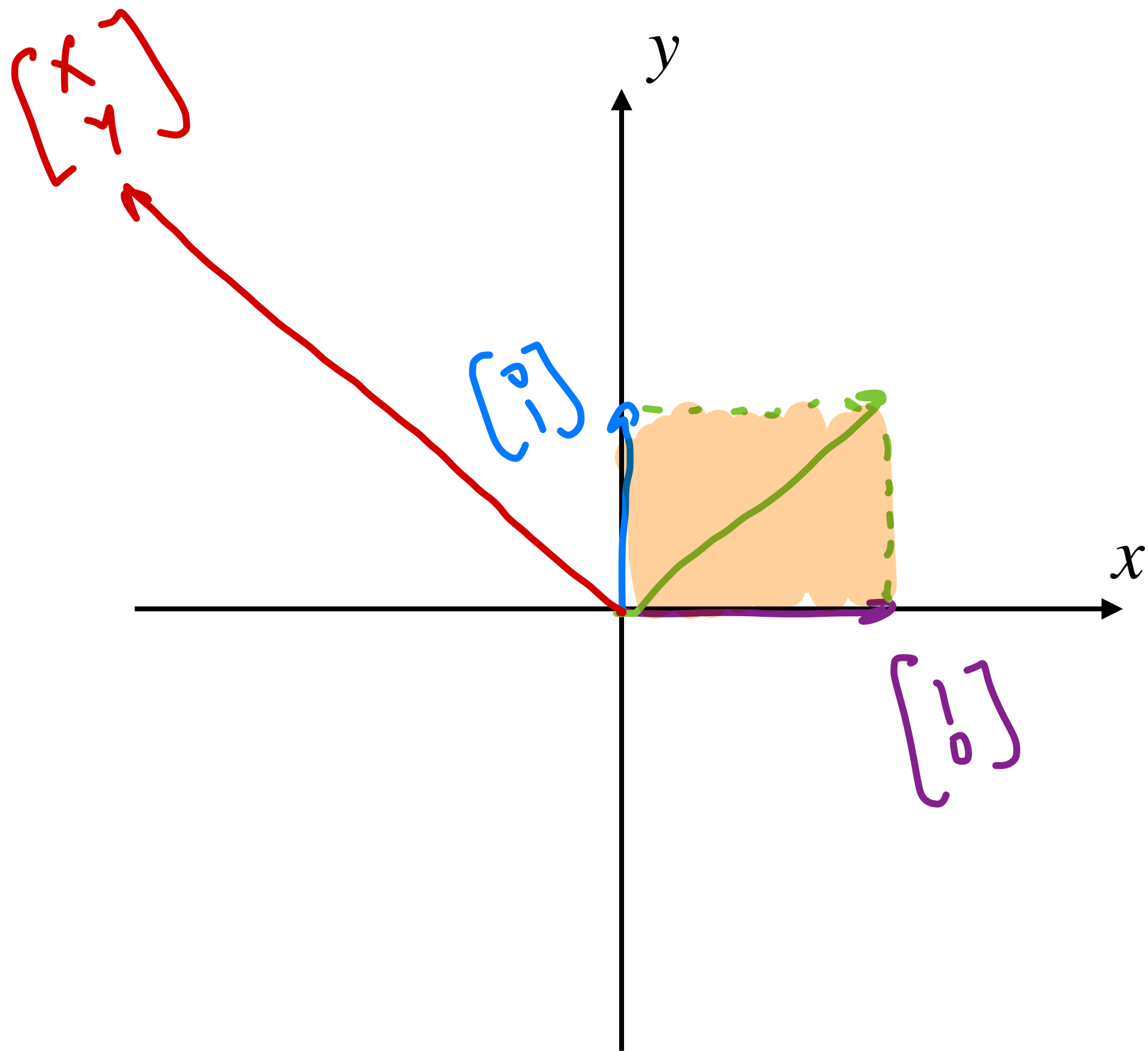


Question

$$\mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$

Draw how this matrix transforms points. What kind of transformation does it represent?

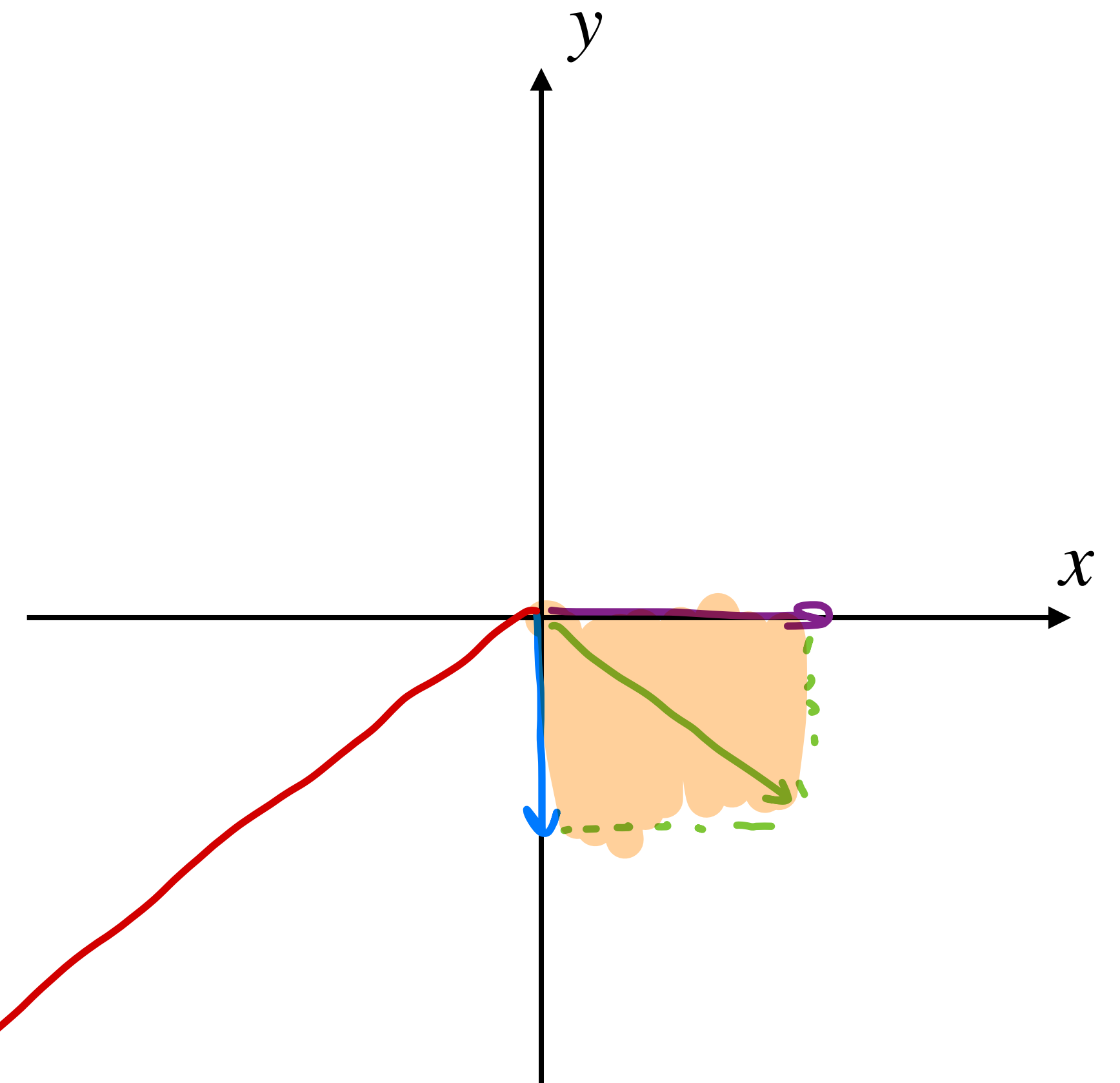
Answer: Reflection



$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$



$$\begin{bmatrix} x \\ -y \end{bmatrix}$$



demo

Matrix of a Linear Transformations

Recall: Linear Transformations

Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

① $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ (additivity)

② $T(c\vec{u}) = cT(\vec{u})$ (homogeneity)

Matrix trans. $\vec{x} \mapsto A\vec{x}$ are linear

Recall: Linear Transformations

Df. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Recall: Linear Transformations

Df. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Matrix transformations are linear transformations

Are there any other kinds of linear transformations?

NO

Matrix of a Linear Transformation

Thm. T is linear if and only if there is a matrix A such that $T(\mathbf{v}) = A\mathbf{v}$. whose corresponding transformation is T (which "implements" T)

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Linear transformations are **exactly**
matrix transformations

Today's Fundamental Concern

Given a linear transformation T , how do we find the matrix A such that $T(\mathbf{v}) = A\mathbf{v}$?

Thought Experiment

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Suppose I am holding a matrix A

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Your objective is to figure out what A is

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But you're only allowed to ask the question:

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But you're only allowed to ask the question:

What is $A\mathbf{v}$?

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(you pick the $\mathbf{v}$'s, and I have to tell the truth)

Thought Experiment

Suppose I am holding a matrix A

Your objective is to figure out what A is

But you're only allowed to ask the question:

What is $A\mathbf{v}$?

(you pick the $\mathbf{v}$'s, and I have to tell the truth)

This is basically linear algebraic battleship

Recall: Matrix-Vector Multiplication

$$\underbrace{\begin{bmatrix} \vec{a}_1 & \dots & \vec{a}_n \end{bmatrix}}_A \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = v_1 \vec{a}_1 + \dots + v_n \vec{a}_n = \sum_{i=1}^n v_i \vec{a}_i$$

$$A \vec{e}_1 = \underbrace{\vec{a}_1}_{\text{first column}}$$

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\begin{aligned} A \vec{e}_1 &= 1 \vec{a}_1 + 0 \vec{a}_2 + \dots + 0 \vec{a}_n \\ &= \vec{a}_1 \end{aligned}$$

$$A \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = \underbrace{\vec{a}_2}_{\text{second column}}$$

Recall: Calculating $A\mathbf{v}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ \vdots \\ ? \end{bmatrix}$$

$$b_1 = a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n = \sum_{i=1}^n a_{1i}v_i$$

Recall: Matrix-Vector Multiplication

Definition. Given a $(m \times n)$ matrix A with columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, and a vector $\mathbf{v}$ in $\mathbb{R}^n$, we define

$$A\mathbf{v} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{a}_1 + v_2\mathbf{a}_2 + \dots v_n\mathbf{a}_n$$

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$A\mathbf{v}$ is a linear combination of the columns of A with weights given by $\mathbf{v}$

Isolating the first column

The Takeaway

We can learn the first column of the matrix

implementing T by looking at $T \begin{pmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \end{pmatrix}$

Standard Basis

$$e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ \vdots \\ n \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \\ \vec{e}_4 \end{matrix}$$

Standard Basis

Df. The *n-dimensional standard basis vectors* (or standard coordinate vectors) are the vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ where

$$\mathbf{e}_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ \vdots \\ i-1 \\ i \\ i+1 \\ \vdots \\ n-1 \\ n \end{matrix}$$

Standard Basis

Def (Alternative). The n -dimensional standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ are the columns of the $n \times n$ identity matrix

$$I = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \dots \quad \mathbf{e}_n]$$

Standard Basis and the Matrix Equation

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{e}_i = \mathbf{a}_i$$

$$(\mathbf{e}_i)_j = \begin{cases} 1 & i=j \\ 0 & \text{o.w.} \end{cases}$$

$$\sum_{j=1}^n \vec{a}_j (\mathbf{e}_i)_j = \vec{a}_i$$

Standard Basis and Vector Coordinates

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \dots + v_n \mathbf{e}_n$$

Column vectors describe how to write a vector as a linear combination of the standard basis

Standard Basis and Linear Transformations

Thm. For a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$, the matrix

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \dots \quad T(\mathbf{e}_n)]$$

is the unique matrix such that $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v}$ in $\mathbb{R}^n$

Formally

$$T(\mathbf{v}) = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix} \mathbf{v}$$

$$T(\vec{v}) =$$

$$T\left(\sum_{i=1}^n v_i \vec{e}_i\right) = \text{(by lin)}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

$$\sum v_i T(\vec{e}_i) = \text{(by mat vec. mul)}$$

$$\begin{bmatrix} T(\vec{e}_1) & \dots & T(\vec{e}_n) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \vec{v}$$

How To: Matrices of Linear Transformations

Q. Find the matrix that implements the transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

A. Determine the images of standard basis vectors under T . The desired matrix is

$$\begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix}$$

Example

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} 5x_1 - 3x_3 \\ 2x_2 + x_3 \end{bmatrix}$$

equiv.

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 5(1) - 3(0) \\ 2(0) + 3(0) \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$\vec{x} \mapsto \begin{bmatrix} 5 & 0 & -3 \\ 0 & 2 & 1 \end{bmatrix} \vec{x}$$

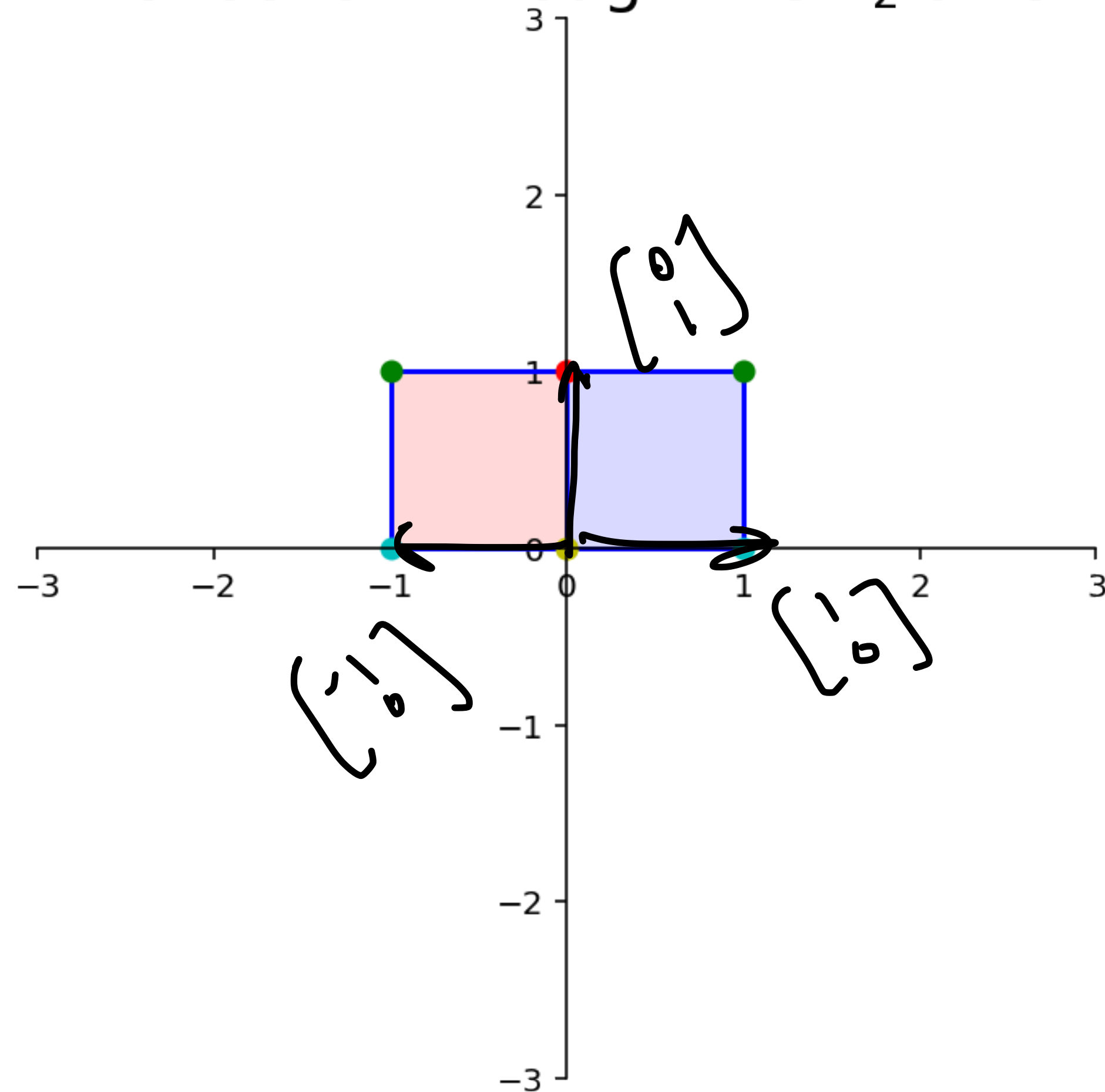
$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & -3 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5x_1 - 3x_3 \\ 2x_2 + x_3 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

Example: Reflection through the x_2 -axis

Reflection through the x_2 axis



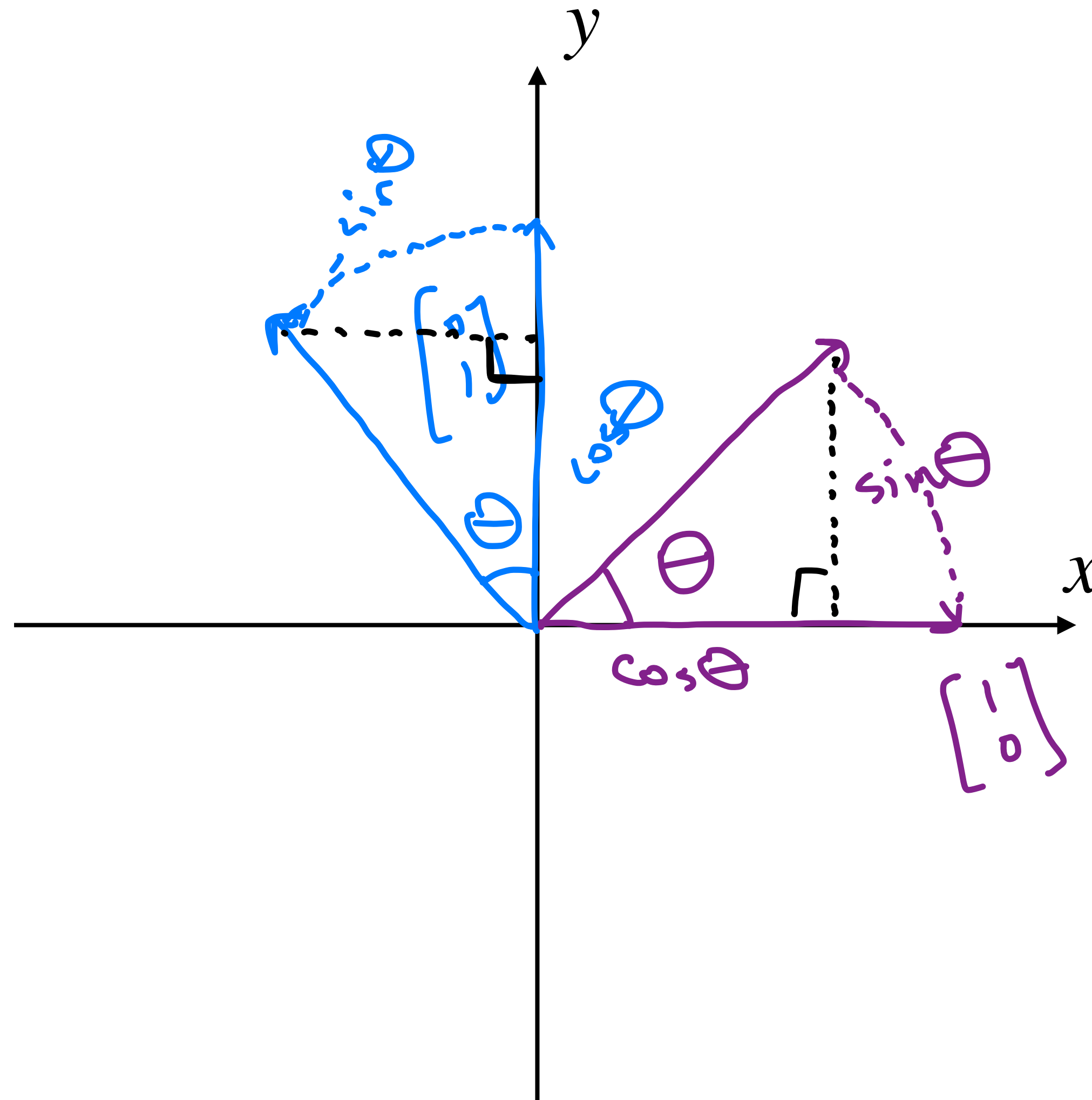
$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\vec{x} \mapsto \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \vec{x}$$

Example: General Rotation

How does rotation affect the standard basis?



$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

Rotation Matrix

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

The Takeaway: We can figure out the matrix that implements a linear transformations by understanding what it does to the standard basis

One-to-One and Onto

Recall: Motivating Questions

What kind of functions can we define in this way?

How do we interpret what the transformation does to a set of vectors?

How does this relate back to matrix equations?

Recall: A New Interpretation of the Matrix Equation

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A
transforms into $\mathbf{b}$?

Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A
transforms into $\mathbf{b}$

Recall: A New Interpretation of the Matrix Equation

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A
transforms into $\mathbf{b}$?

Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A
transforms into $\mathbf{b}$

What about other questions?

Other Questions Like...

Does $A\mathbf{x} = \mathbf{b}$ have a solution for any choice of $\mathbf{b}$?
(full span)

Does $A\mathbf{x} = \mathbf{0}$ have a unique solution?

(lin. ind.)

Other Questions Like...

Does $A\mathbf{x} = \mathbf{b}$ have at least one solution for any choice of $\mathbf{b}$?

Does $A\mathbf{x} = \mathbf{b}$ have at most one solution for any choice of $\mathbf{b}$?

Verify

①

$A\mathbf{x} = \mathbf{0}$ has a
unique solution

$\equiv$

②

$A\mathbf{x} = \mathbf{b}$ has at most one
solution

① $\Rightarrow$ ② by not ② $\Rightarrow$ not ①

Suppose $A\vec{v}_1 = \vec{b}$ $A\vec{v}_2 = \vec{b}$ $\vec{v}_1 \neq \vec{v}_2$

$$A(\vec{v}_1 - \vec{v}_2) = A\vec{v}_1 - A\vec{v}_2 = \vec{b} - \vec{b} = \vec{0}$$

$\neq 0$ $\vec{v}_1 - \vec{v}_2$ is nontrivial sol.

② $\Rightarrow$ ① Challenge.

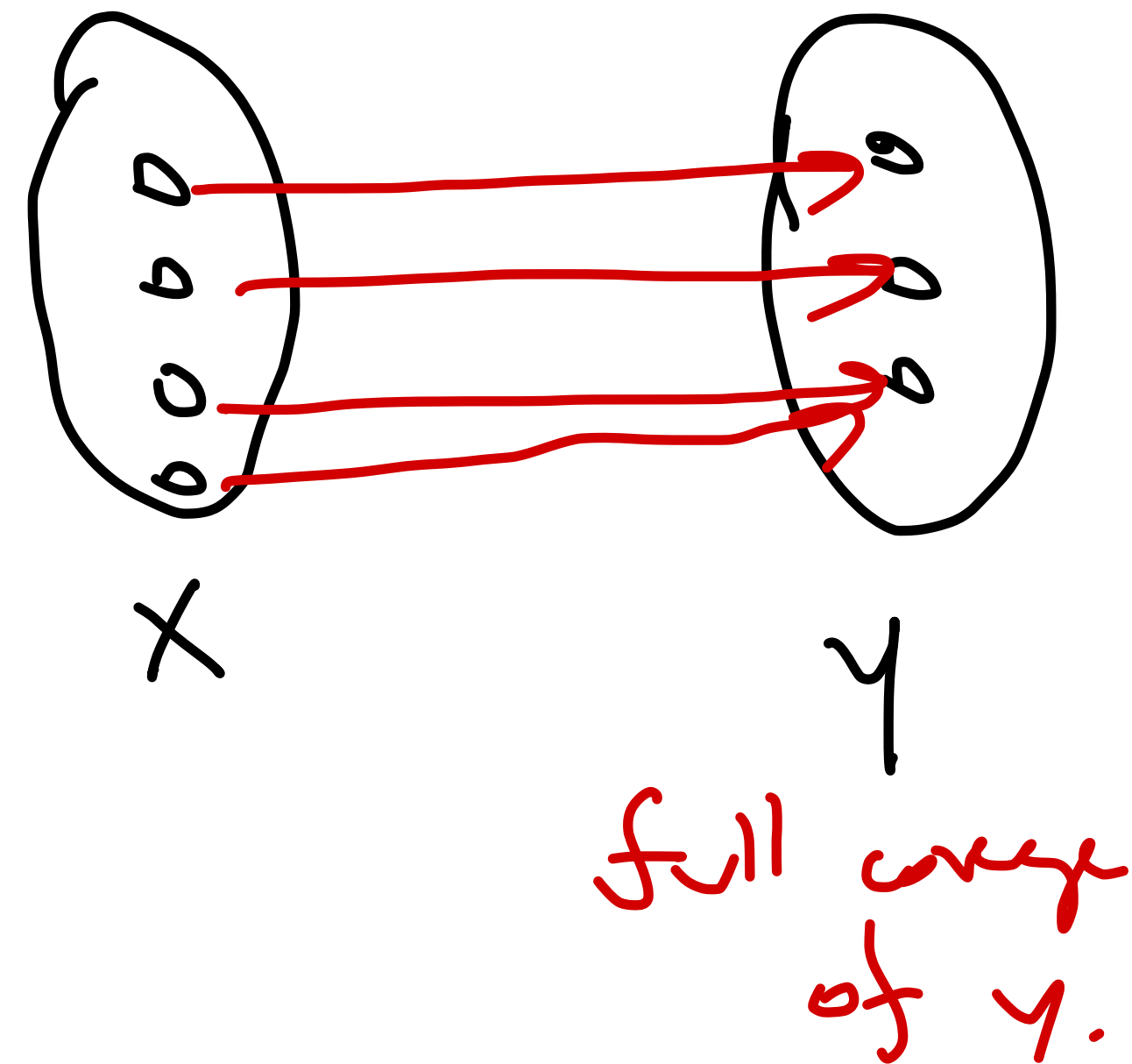
Onto

~~Df.~~ $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is onto

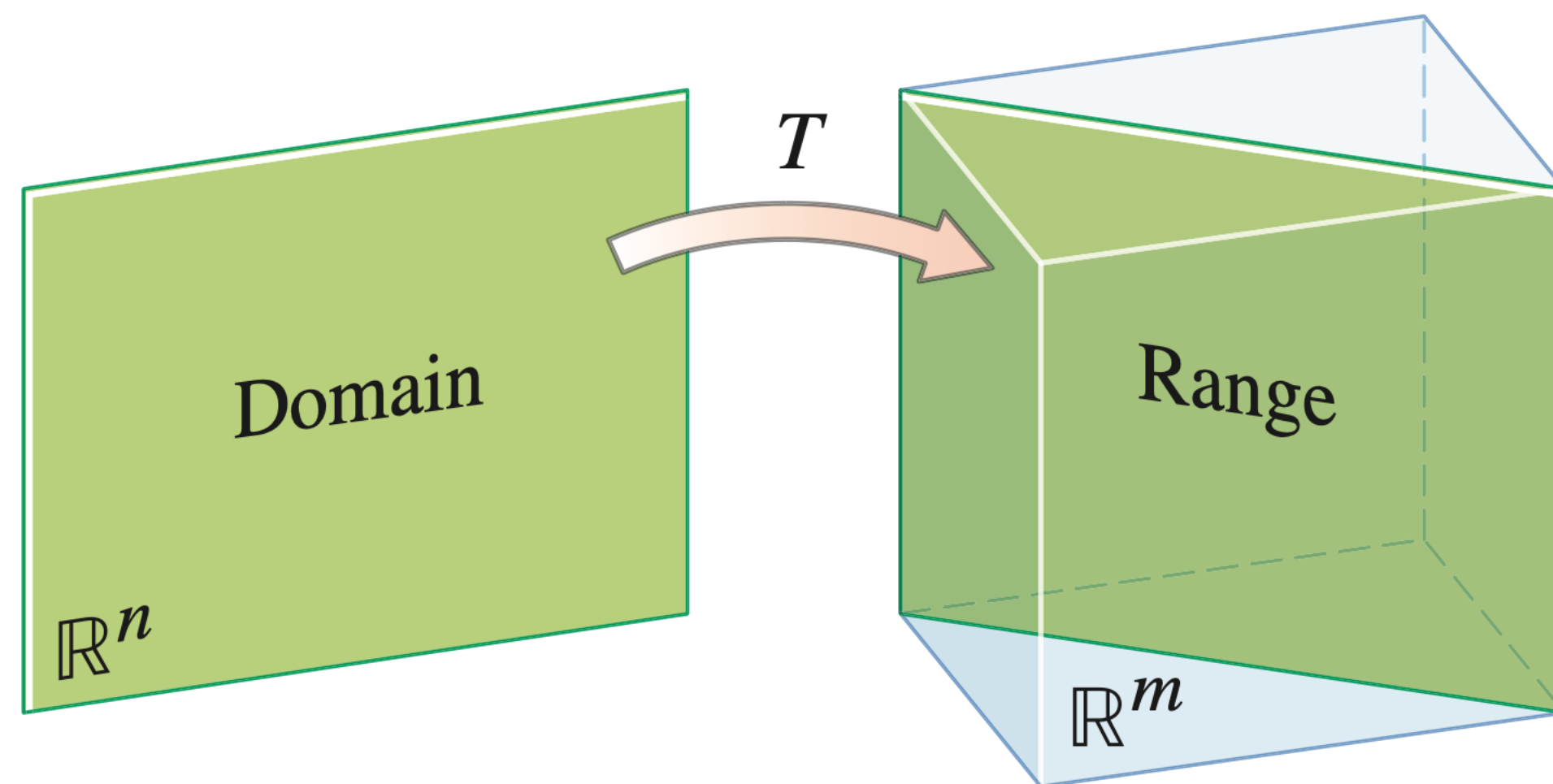
if for every $\vec{v} \in \mathbb{R}^m$
there is at least one $\vec{u} \in \mathbb{R}^n$

s.t. $T(\vec{u}) = \vec{v}$

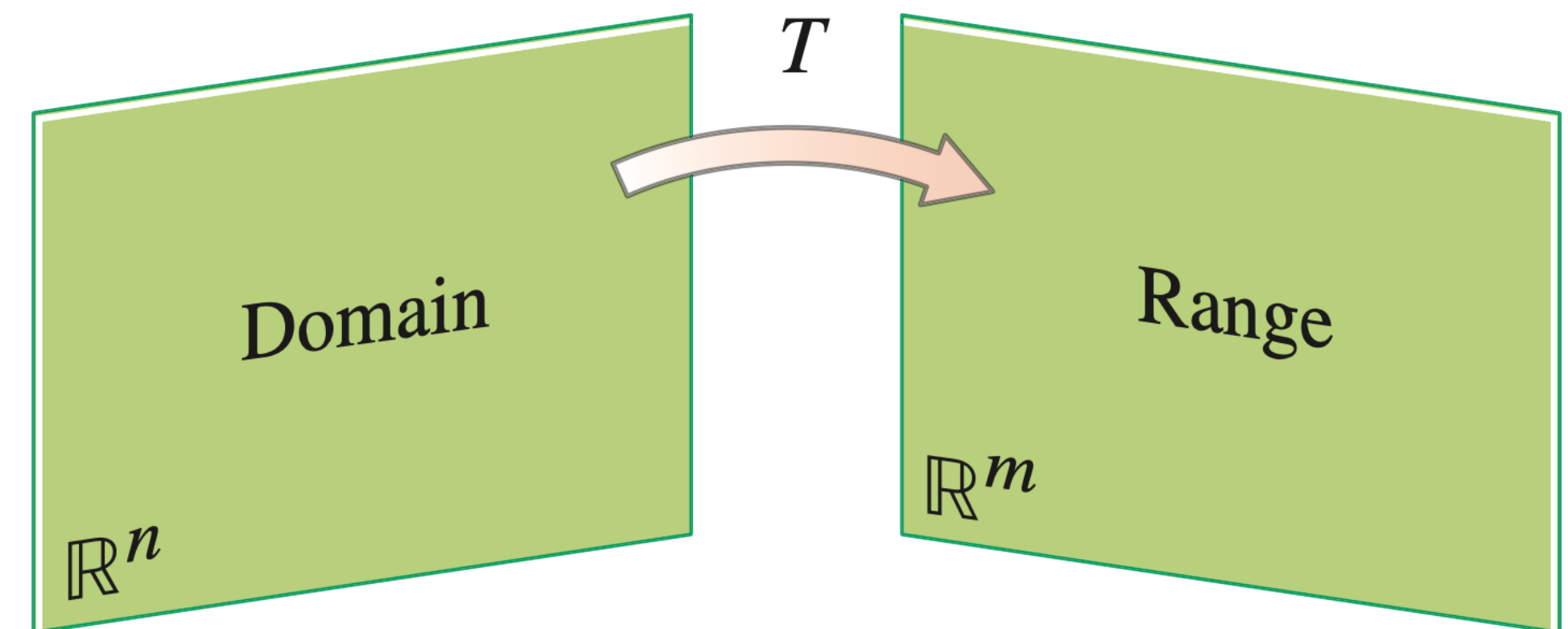
preimage of $\vec{v}$



Onto (Pictorially)

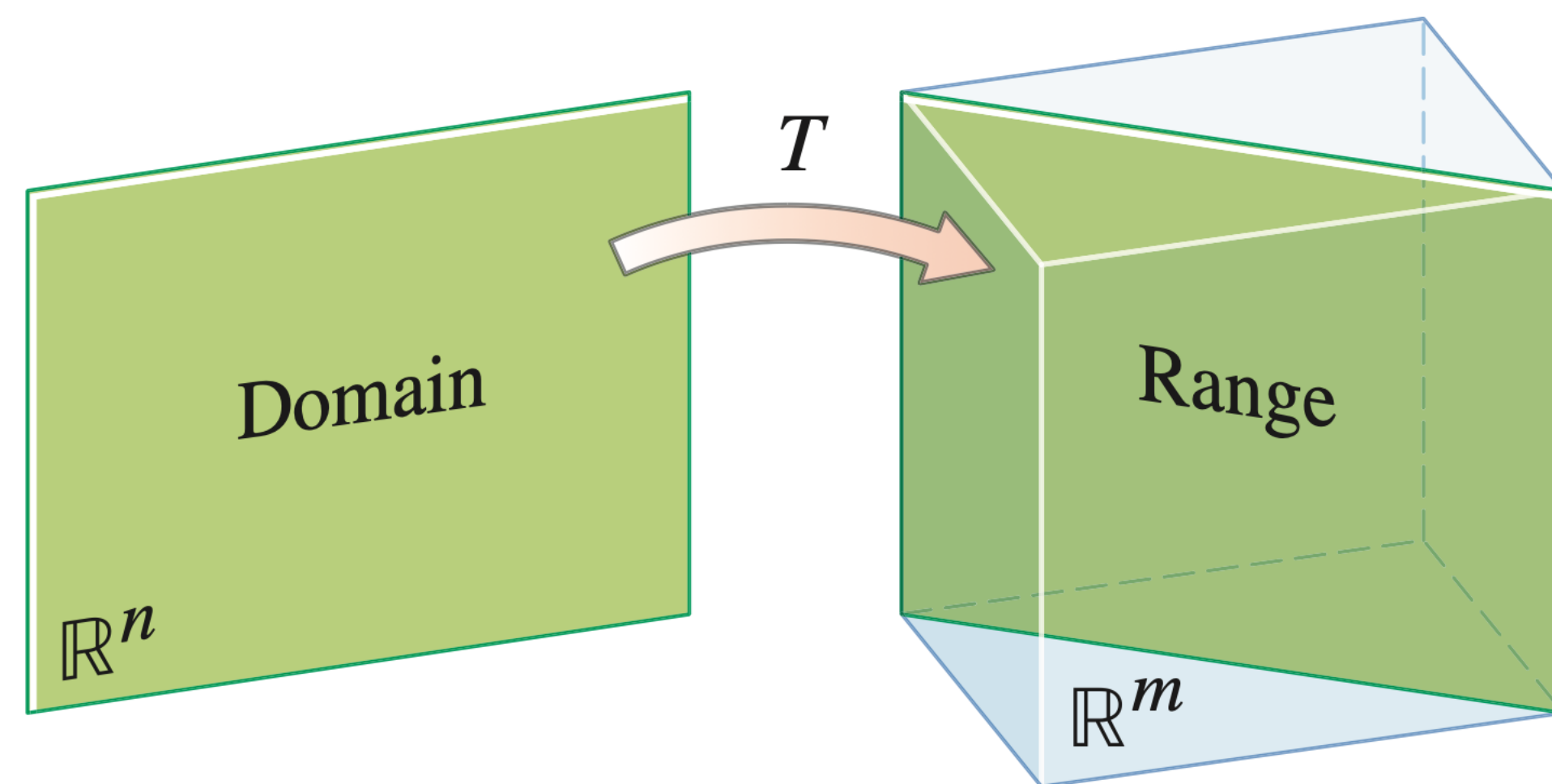


T is *not* onto $\mathbb{R}^m$

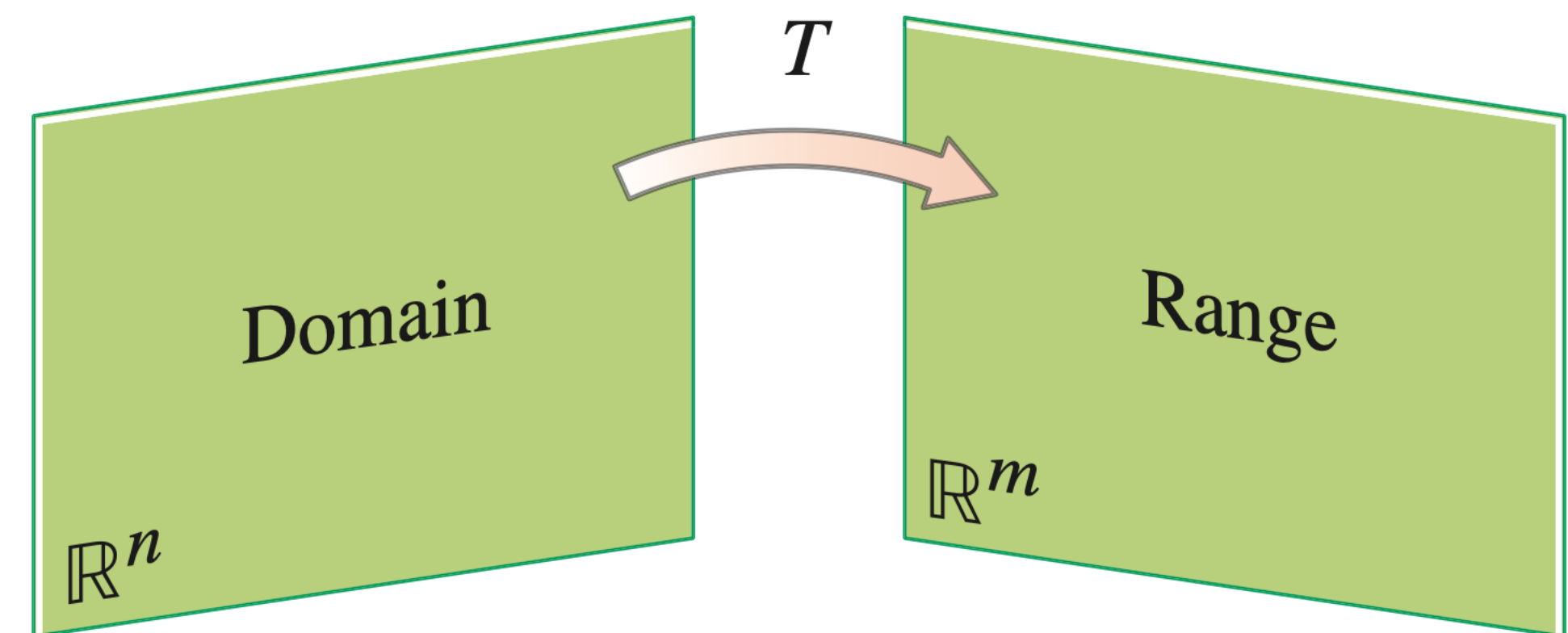


T is onto $\mathbb{R}^m$

Onto (Pictorially)



T is *not* onto $\mathbb{R}^m$



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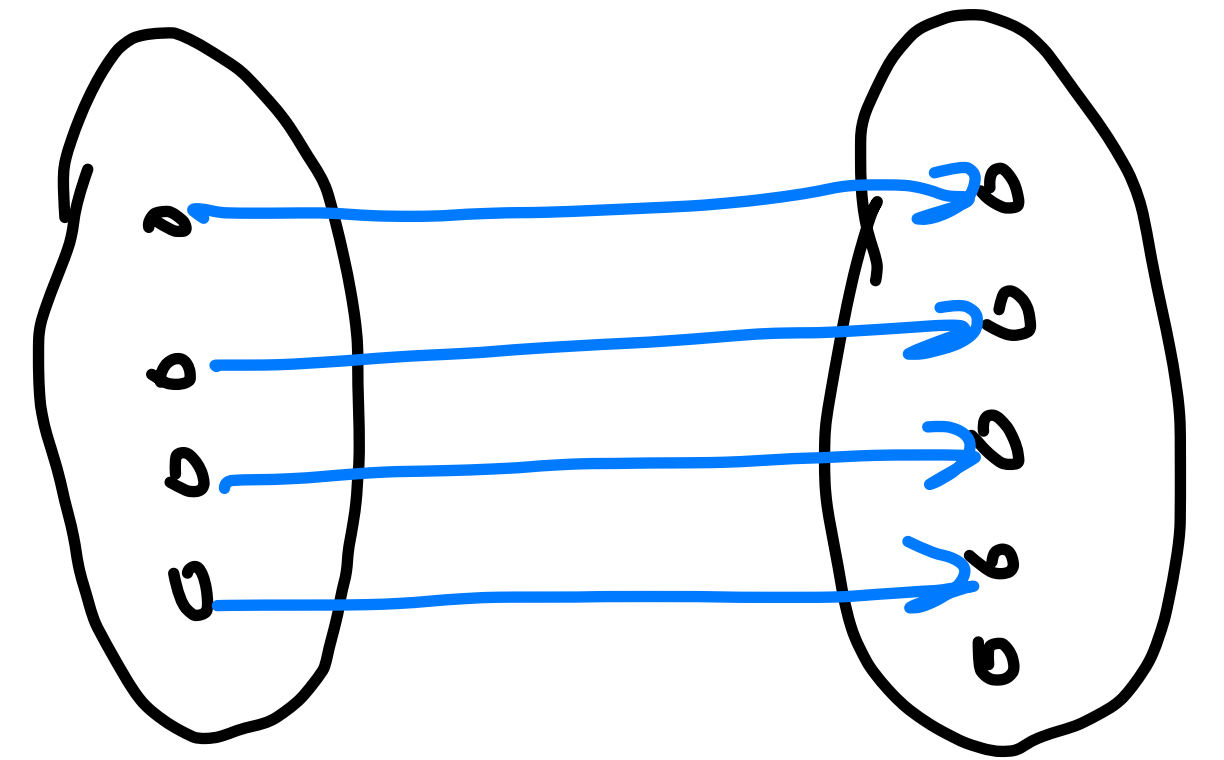
T is onto if its range = its codomain

One-to-One

Def. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is 1-1

if for every $\vec{w} \in \mathbb{R}^m$ there is
at most one $\vec{v} \in \mathbb{R}^n$ s.t.

$$T(\vec{v}) = \vec{w}$$

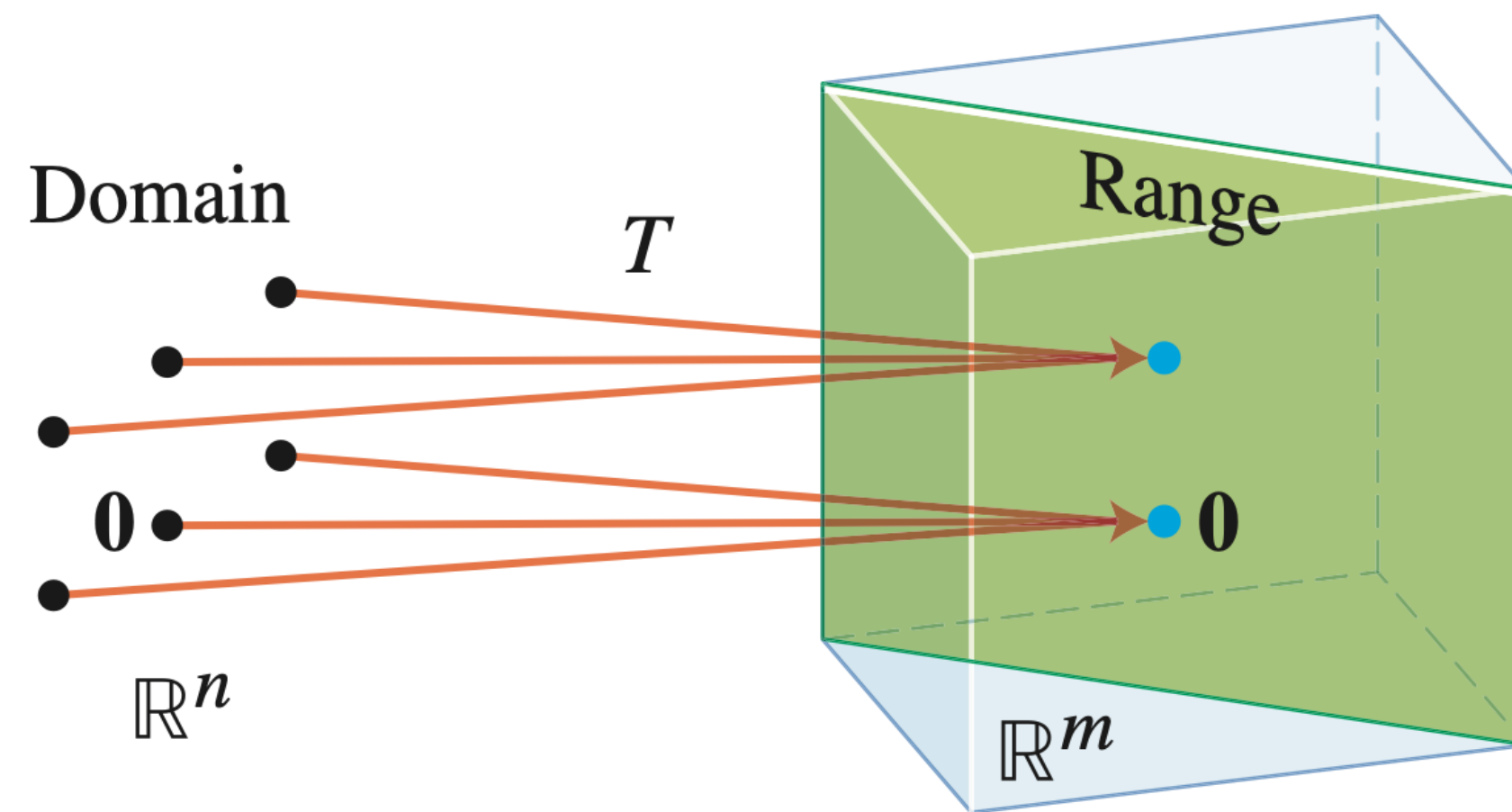


X

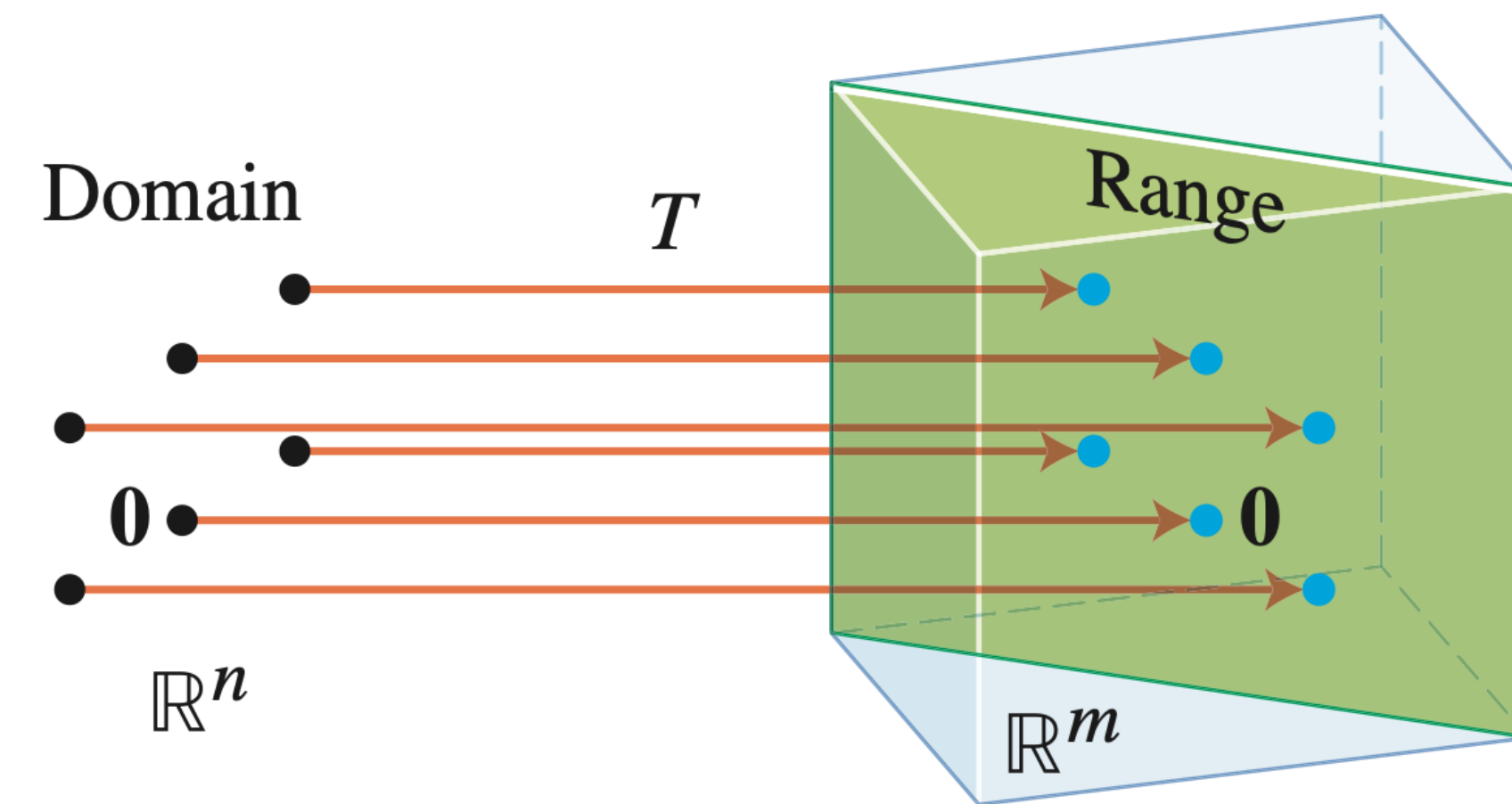
Y

no collisions
in Y.

One-to-One (Pictorially)



T is *not* one-to-one



T is one-to-one

Onto and One-to-One

Onto and One-to-One

Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *onto* if any $\mathbf{b} \in \mathbb{R}^m$ is the image of at least one vector $\mathbf{v} \in \mathbb{R}^n$ (i.e., $T(\mathbf{v}) = \mathbf{b}$)

Onto and One-to-One

Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** if any $\mathbf{b} \in \mathbb{R}^m$ is the **image of at least one vector** $\mathbf{v} \in \mathbb{R}^n$ (i.e., $T(\mathbf{v}) = \mathbf{b}$)

Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** if any $\mathbf{b} \in \mathbb{R}^m$ is the **image of at most one vector** $\mathbf{v} \in \mathbb{R}^n$

Taking Stock: Onto

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Thm. The following are logically equivalent for
 $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

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 $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is onto

Taking Stock: Onto

Thm. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is onto

» $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$

Taking Stock: Onto

Thm. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is onto
- » $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$
- » $\text{range}(T) = \text{codomain}(T)$

Taking Stock: Onto

Thm. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is onto
- » $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$
- » $\text{range}(T) = \text{codomain}(T)$
- » the columns of A span $\mathbb{R}^m$

Taking Stock: Onto

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- » $\text{range}(T) = \text{codomain}(T)$
- » the columns of A span $\mathbb{R}^m$
- » A has a pivot position in every row

Taking Stock: One-to-One

Taking Stock: One-to-One

Theorem. The following are logically equivalent
for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

Taking Stock: One-to-One

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for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is one-to-one

Taking Stock: One-to-One

Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is one-to-one

» $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$

Taking Stock: One-to-One

Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is one-to-one
- » $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
- » $A\mathbf{x} = \mathbf{0}$ has only the trivial solution

Taking Stock: One-to-One

Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is one-to-one
- » $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
- » $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
- » The columns of A are linearly independent

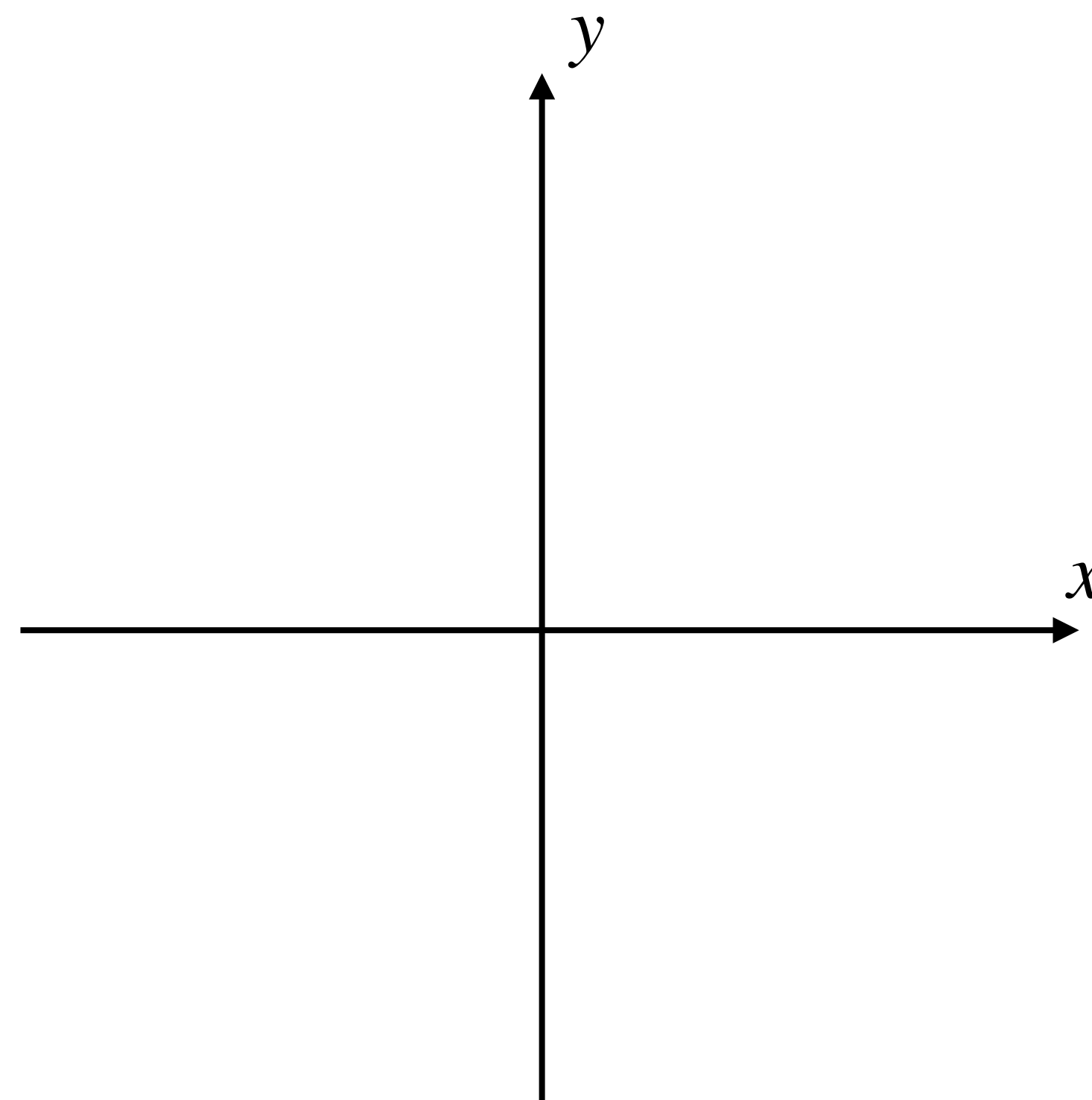
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- » A has a pivot position in every column

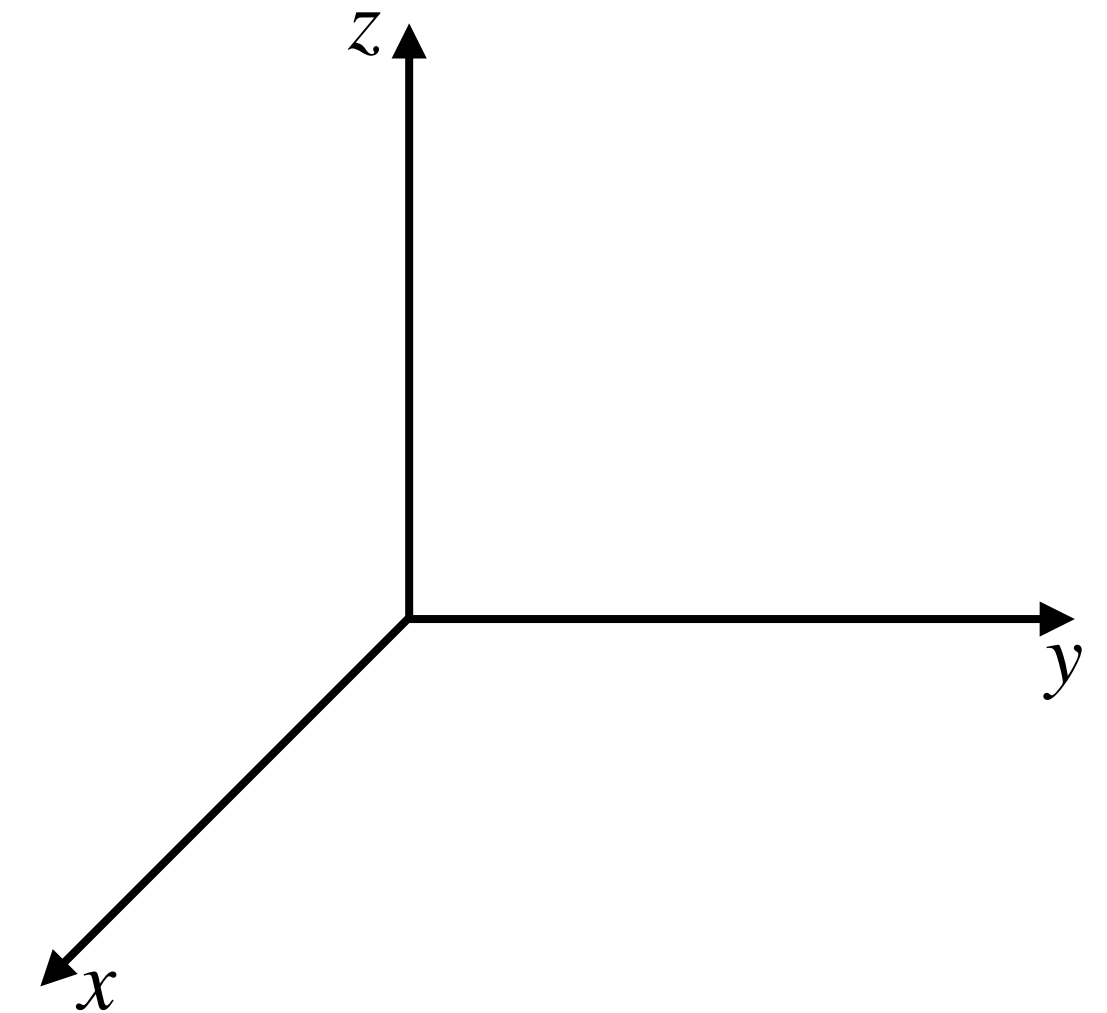
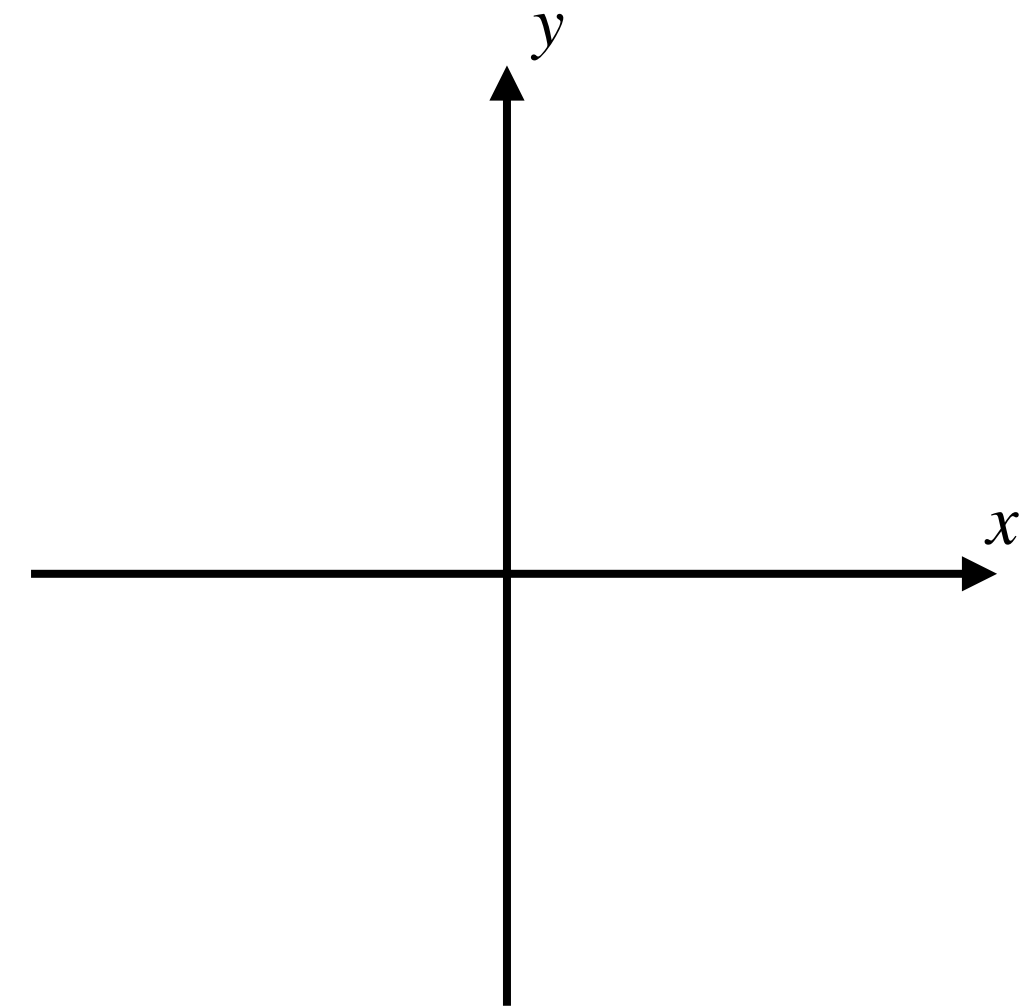
Example: both 1-1 and onto

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$



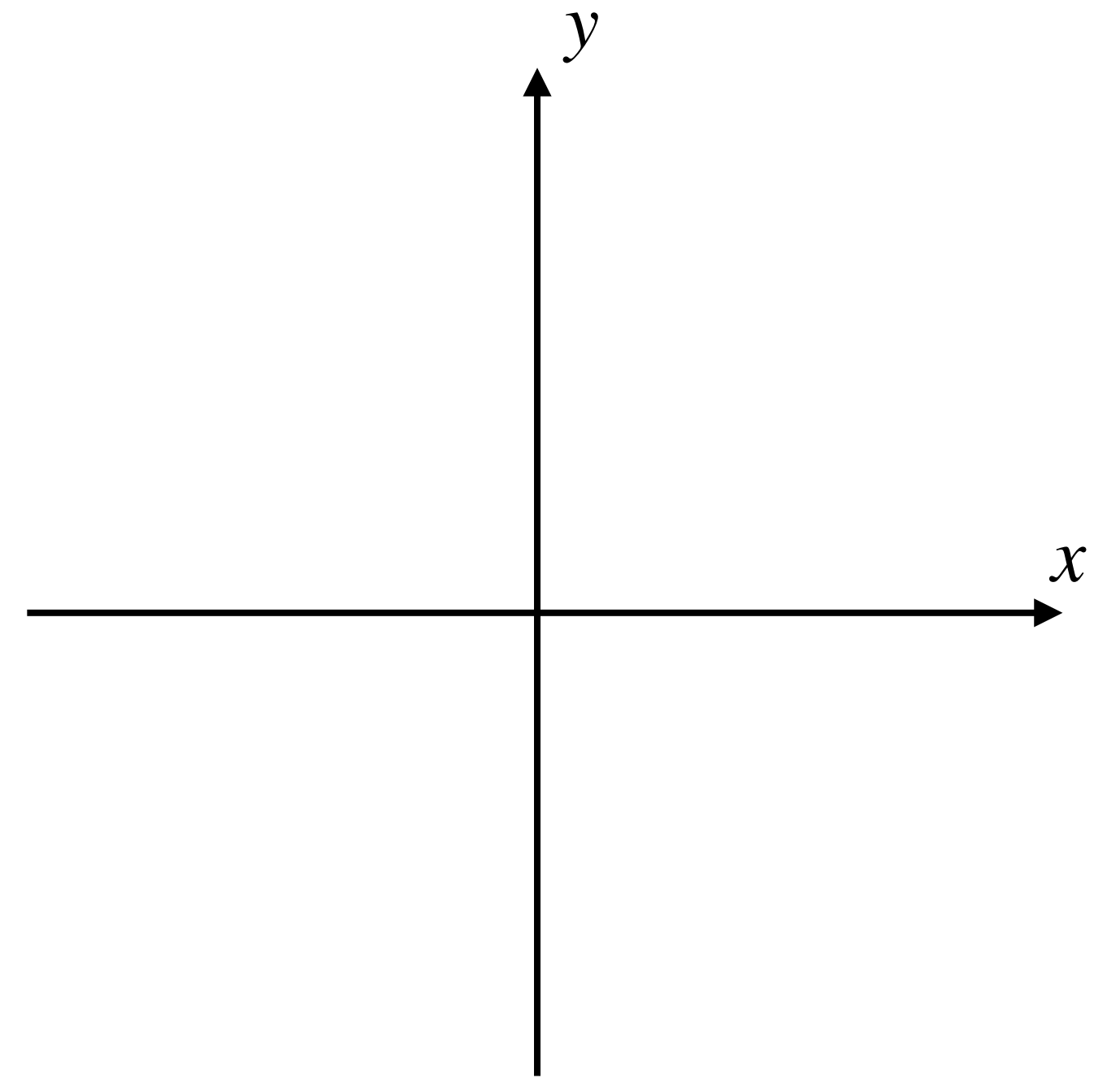
Example: 1-1, not onto

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}$$



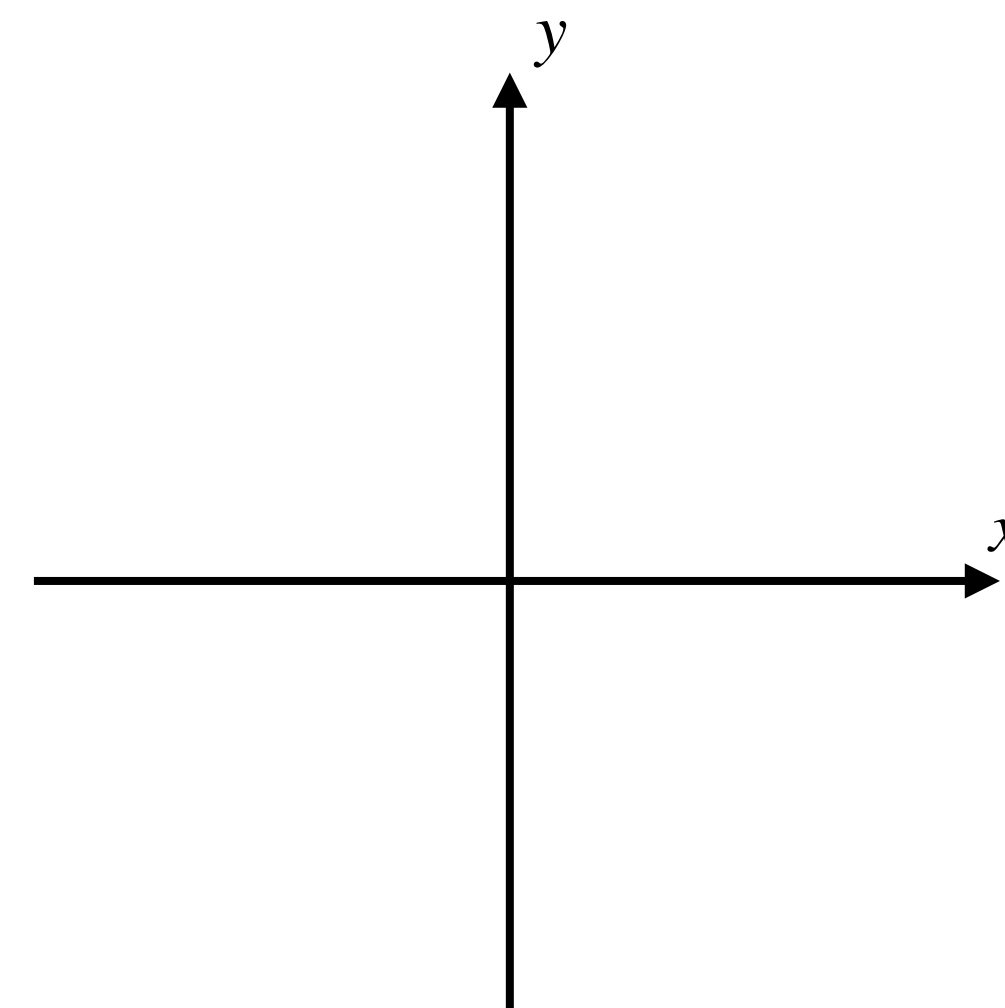
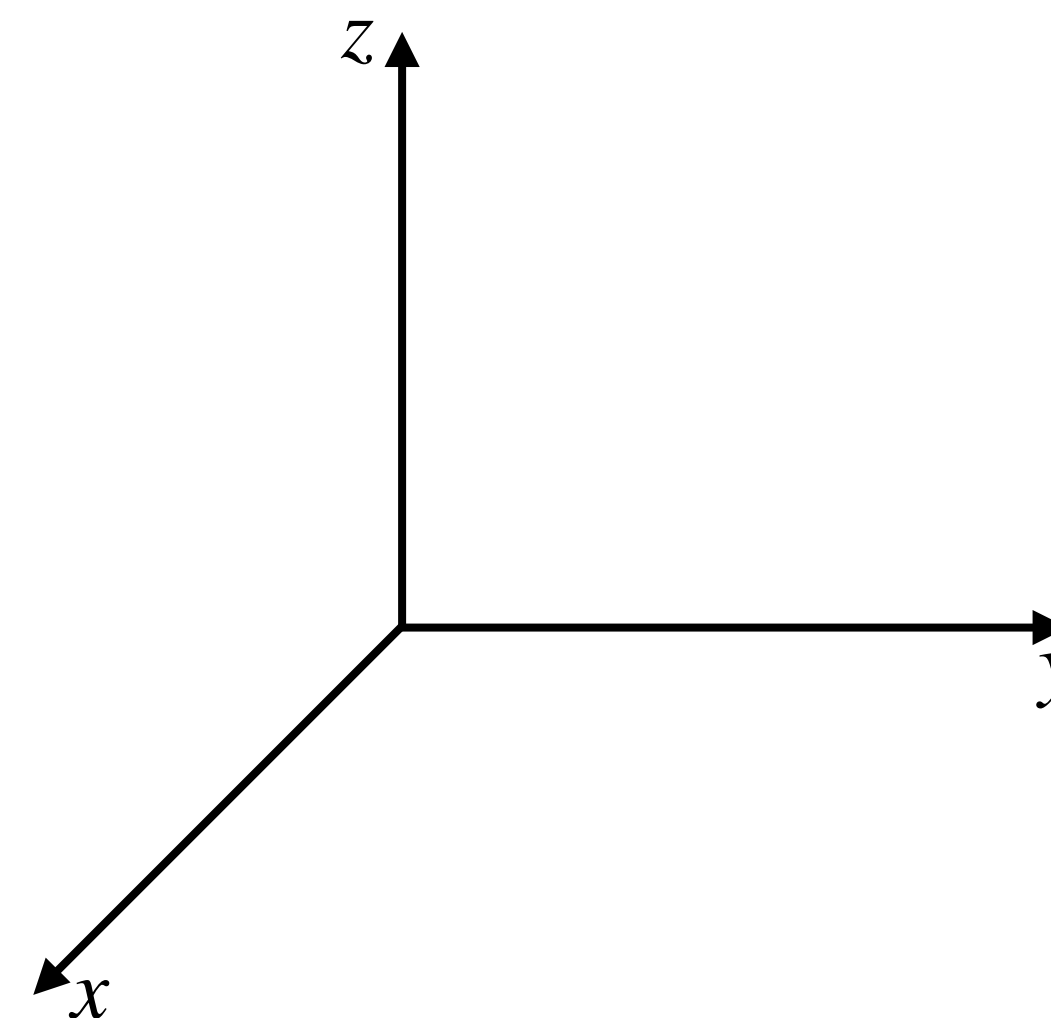
Example: not 1-1, not onto

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$



Example: onto, not 1-1

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



Summary

Matrix transformations and linear transformations are **the same thing**

We can find these matrices by looking at how the transformation behaves on the **standard basis**

We can reason about matrix equations by directly reasoning about the linear transformations